

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Sw Testing

Section Id :	640653141349
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	17
Number of Questions to be attempted :	17
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653398272
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532166074 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SOFTWARE TESTING (COMPUTER

BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406537043120.  Yes

6406537043121.  No

Sub-Section Number :

2

Sub-Section Id :

640653398273

Question Shuffling Allowed :

Yes

Question Number : 2 Question Id : 6406532166075 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

In an organization, software teams understand that the objective of testing is to reduce risks. Due to this, the developers and testers are cooperating with each other to help fix issues and reduce risks. Considering this scenario, identify the minimum process maturity level of this organization.

Options :

6406537043122.  Level 1

6406537043123.  Level 2

6406537043124.  Level 3

6406537043125.  Level 4

Question Number : 3 Question Id : 6406532166076 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

An e-commerce application contains separate modules for user authentication, shopping cart management, and payment processing. Although each module behaves correctly in isolation, the engineering team discovers issues when customer login sessions are transferred incorrectly between modules during checkout.

Which of the following types of testing is primarily intended to identify such problems?

Options :

6406537043126.  Stress testing

6406537043127.  Integration testing

6406537043128.  Regression testing

6406537043129.  Usability testing

Question Number : 4 Question Id : 6406532166077 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following statements and choose the correct option.

- i) In JUnit, each test is embedded into one test method.
ii) The assertion `assertFalse(5 % 2 == 0)` would fail.

Options :

6406537043130. ✓ Statement i) is correct and Statement ii) is incorrect.
6406537043131. ✗ Statement ii) is correct and Statement i) is incorrect.
6406537043132. ✗ Both statements are incorrect.
6406537043133. ✗ Both statements are correct.

Question Number : 5 Question Id : 6406532166078 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

A _____ is a software component or test tool that replaces a component that takes care of the control and/or the calling of a software component.

Fill in the blank with correct option.

Options :

6406537043134. ✗ test method
6406537043135. ✗ test suite
6406537043136. ✓ test driver
6406537043137. ✗ test stub

Question Number : 6 Question Id : 6406532166079 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

A user is browsing a website. She types a query in a text box and clicks on Search. Upon clicking the Search button, a request is made to the server which then retrieves the relevant items and returns a response. Which of the following types of interfaces does this scenario represent?

Options :

6406537043138. ✗ Procedure call interface
6406537043139. ✗ Shared memory interface
6406537043140. ✗ External file interface
6406537043141. ✓ Message passing interface

Question Number : 7 Question Id : 6406532166080 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following statements regarding structural graph coverage criteria, and choose the correct option.

- i) Prime path coverage always subsumes edge-pair coverage.
- ii) A prime path is also a simple path.
- iii) Complete path coverage may be feasible for some CFGs.

Options :

6406537043142. ✖ Statement i) and iii) are correct. Statement ii) is incorrect.

6406537043143. ✔ Statement ii) and iii) are correct. Statement i) is incorrect.

6406537043144. ✖ Statement i) and ii) are correct. Statement iii) is incorrect.

6406537043145. ✖ Statement i) is correct. Statement ii) and iii) are incorrect.

Question Number : 8 Question Id : 6406532166081 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider a control flow graph G . In which of the following types of testing, the number of test requirements (TRs) is equal to the cyclomatic complexity of G ?

Options :

6406537043146. ✔ Basis Path Testing

6406537043147. ✖ Prime Path Coverage

6406537043148. ✖ Data Flow Testing

6406537043149. ✖ Edge-Pair Coverage

Question Number : 9 Question Id : 6406532166082 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Which of the following models is best to capture state behavior when testing sequencing constraints of a program or module?

Options :

6406537043150. ✖ Activity Diagrams

6406537043151. ✖ Control Flow Graphs (CFGs)

6406537043152. ✔ Finite State Machines (FSMs)

6406537043153. ✖ Call Graphs

Question Number : 10 Question Id : 6406532166083 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Which of the following graph coverage criteria is equivalent to transition coverage on a finite state machine (FSM)?

Options :

6406537043154. ✖ Node coverage

6406537043155. ✔ Edge coverage

6406537043156. ✖ Edge-pair coverage

6406537043157. ✖ Prime path coverage

Question Number : 11 Question Id : 6406532166084 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Which of the following options correctly describes All-Uses Coverage criterion for a variable x ?

Options :

6406537043158. ✖ At least one definition of x must reach at least one use.

6406537043159. ✖ Each definition of x must reach at least one possible use.

6406537043160. ✖ At least one definition of x must reach every possible use.

6406537043161. ✔ Each definition of x must reach every possible use.

Sub-Section Number :

3

Sub-Section Id :

640653398274

Question Shuffling Allowed :

Yes

Question Number : 12 Question Id : 6406532166085 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements are correct regarding the representation of graphs?

Options :

6406537043162. ✔ Adjacency list representation provides a compact way to represent sparse graphs.

6406537043163. ✖ Adjacency matrix representation cannot be used for control flow graphs.

6406537043164. ✖ Adjacency list representation cannot be used for control flow graphs.

6406537043165. ✔ Adjacency matrix representation provides a compact way to represent dense graphs.

Question Number : 13 Question Id : 6406532166086 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following types of testing methods are *black box*, i.e. those that do not require knowledge about the internals of the program?

Options :

6406537043166. ✖ Integration Testing

6406537043167. ✖ Unit Testing

6406537043168. ✔ Beta Testing

6406537043169. ✔ Stress Testing

Question Number : 14 Question Id : 6406532166087 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements are true regarding a decision-to-decision (DD) path?

Options :

6406537043170. ✖ Every chain is also a valid DD-path.

6406537043171. ✔ It can be a single vertex with out-degree ≥ 2 .

6406537043172. ✖ It cannot consist of only the final vertex.

6406537043173. ✔ It can consist of only the initial vertex.

Question Number : 15 Question Id : 6406532166088 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the following Java code for `MathUtils` class, there are some faults in it. The method `sumPositiveNumbers(int[] arr)` needs to be tested using the JUnit test class `TestPositiveSum`.

```

public static class MathUtils {
    // Inputs: An array of integers.
    // Effects: Computes and returns the sum of all numbers greater than 0.
    // Assumptions: Returns 0 if the array is null or empty.
    public static int sumPositiveNumbers(int[] arr) {
        if (arr == null || arr.length == 0) {
            return 0;
        }
        int sum = 0;
        for (int i = 0; i < arr.length - 1; i++) {
            if (arr[i] > 0) {
                sum += arr[i];
            }
        }
        return sum;
    }
}

/* ----- Test Class ----- */
import static org.junit.Assert.*;
import org.junit.Test;

public class TestPositiveSum {
    @Test
    public void testCase1() {
        int[] arr = { 2, 3, 4, 5 };
        int result = MathUtils.sumPositiveNumbers(arr);
        assertEquals(14, result);
    }

    @Test
    public void testCase2() {
        int[] arr = { 10, 20, -5, -1 };
        int result = MathUtils.sumPositiveNumbers(arr);
        assertEquals(30, result);
    }

    @Test
    public void testCase3() {
        int[] arr = { -1, -2, -3, 0 };
        int result = MathUtils.sumPositiveNumbers(arr);
        assertEquals(0, result);
    }

    @Test
    public void testCase4() {
        int[] arr = { -4, -2, 8, 10 };
        int result = MathUtils.sumPositiveNumbers(arr);
        assertEquals(18, result);
    }
}

```

Identify which of the test cases would be able to uncover the faults in the program.

Options :

6406537043174. ✓ `testCase1()`

6406537043175. ✗ `testCase2()`

6406537043176. ✗ `testCase3()`

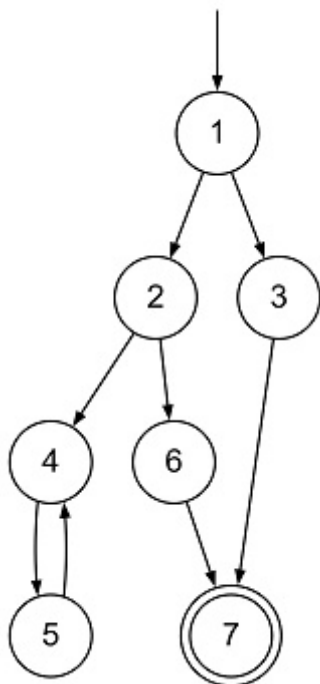
6406537043177. ✓ `testCase4()`

Sub-Section Number : 4
Sub-Section Id : 640653398275
Question Shuffling Allowed : No

Question Id : 6406532166089 **Question Type :** COMPREHENSION **Sub Question Shuffling Allowed :** No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix
Question Numbers : (16 to 18)

Question Label : Comprehension

Consider the following control flow graph (CFG), and answer the subsequent questions.



Sub questions

Question Number : 16 **Question Id :** 6406532166090 **Question Type :** MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

How many test requirements are there for edge-pair coverage?

Options :

6406537043178. ✖ 8

6406537043179. ✖ 11

6406537043180. ✓ 7

6406537043181. ✖ 9

Question Number : 17 **Question Id :** 6406532166091 **Question Type :** MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

How many test requirements are there for prime path coverage?

Options :

6406537043182. ✖ 9

6406537043183. ✖ 12

6406537043184. ✖ 7

6406537043185. ✔ 5

Question Number : 18 Question Id : 6406532166092 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

What is the cyclomatic complexity of the given CFG?

Options :

6406537043186. ✔ 3

6406537043187. ✖ 4

6406537043188. ✖ 5

6406537043189. ✖ 6

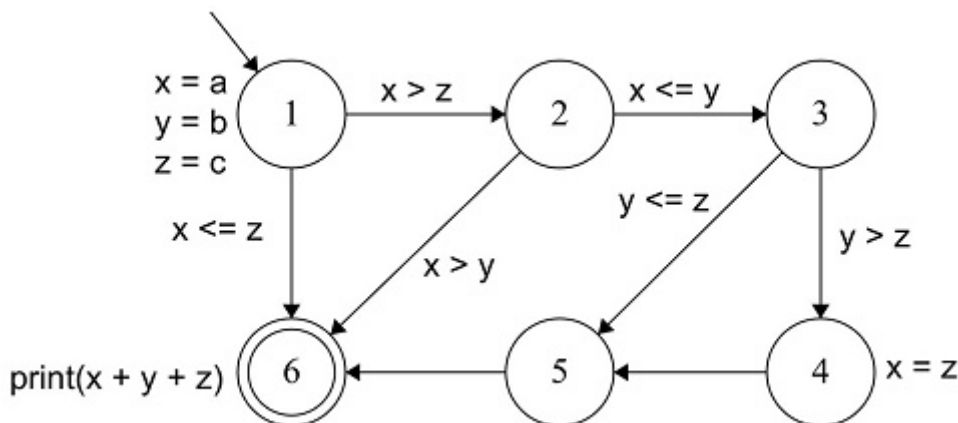
Question Id : 6406532166093 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (19 to 21)

Question Label : Comprehension

Consider the following data flow graph, and answer the subsequent questions.



Sub questions

Question Number : 19 Question Id : 6406532166094 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

How many test requirements are there for All-Defs-Coverage for variable x ?

Options :

6406537043190. ✖ 4

6406537043191. ✔ 2

6406537043192. ✖ 3

6406537043193. ✖ 1

Question Number : 20 Question Id : 6406532166095 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

How many test requirements are there for All-Uses-Coverage for variable x ?

Options :

6406537043194. ✖ 4

6406537043195. ✖ 6

6406537043196. ✔ 5

6406537043197. ✖ 7

Question Number : 21 Question Id : 6406532166096 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

How many unique du-paths are there for variable x ?

Note: Do not consider subsumption for the du-paths.

Options :

6406537043198. ✔ 7

6406537043199. ✖ 6

6406537043200. ✖ 9

6406537043201. ✖ 8